

Group Project 2

Analyzing Recycling Behavior at NUS

DBA5101 Analytics in Managerial Economics

Ng Yu Fei (A0252106U), Tan Hua Swen (A0216984R),
Teo Boon Khai, Marcus (A0167321L)

1. Introduction

In this analysis, we evaluate whether two sequential interventions at NUS, namely shaped openings and informational banners, reduced contamination rates of recycling bins in NUS. The researchers ran a three-phase trial at recycling bins in two sites: Engine and UTown. Recycling bins at Engine were unchanged throughout the 3 phases and served as the control group. Phase 1 at UTown was the baseline, and no intervention was made, Phase 2 involved installing shaped openings at bins and Phase 3 involved placing informational banners. Outcomes were recorded as contamination rates for paper, plastic, and cans at both sites.

Randomized Controlled Trial was not possible in this experiment due to inherent issues with the experiment goals. Firstly, experimenter-controlled randomization was not possible as it was infeasible to randomly assign students to different bins or vice versa. Secondly, students and staff move between Engine and UTown, they could observe intervention at one location and change behavior across locations. These two reasons lead to failure of the independence assumption and stable unit treatment value assumption (SUTVA) needed for RCTs. Instead we chose Difference in Difference (DID) analysis as our causal identification strategy.

2. Goals of analysis

Our goal is to estimate the causal effect of shaped openings and informational banners on recycling behaviour.

3. DID analysis

3.1 Defining Hypothesis

We define the hypothesis for our interventions as follows:

H0: intervention has no effect on contamination rates of paper, plastic and cans

H1: intervention reduces contamination rates of paper, plastic and cans

3.2 Data pre-processing

Our initial dataset consists of contamination data for paper, plastics and cans across dates at Engine and UTown. We carried out some data cleaning: data from 2020-02-02 were removed due to missing observations in both Engine and UTown.

In Phase 1, while both groups had the same number of timepoints, UTown had two timepoints not in Engine ('2020-01-13' and '2020-01-14') while Engine had two timepoints not in UTown ('2020-01-16' and '2020-02-07'). There was also an extra time point recorded for Engine on 2020-03-10 during Phase 3. We have dropped these timepoints to remove imbalance in our dataset. Unbalanced data can introduce bias into our estimator as we cannot be certain that unobserved time points are not systematically related to treatment or outcome.

3.3 Parallel Trends Assumption

We plot a scatter plot for Engine and UTown across phases 1, 2 and 3 and do not notice an immediately observable parallel trend (see figure 3). Next, we plot a line of best fit for Engine and UTown in Phase 1 (figure 4) and notice a visually parallel trend between Engine and UTown across the contaminant types.

We decided to further test this assumption using the **Statistical Test for Pre-Treatment Trend Differences** (figure 5) on Phase 1 to measure the statistical difference between the two groups. After regression on Phase 1 data, we notice that β_2 for PaperContaminant and CanContaminant was 0.4573 and 1.6661 respectively with p-value 0.000. Whereas only PlasticContaminant had a non-significant p-value (0.053). We can infer that PaperContaminant and CanContaminant trends for Engine and UTown were not following a parallel trend and only PlasticContaminant might be following a parallel trend. The implication of this will be discussed when interpreting the results.

3.4 Initial DID Regression

To understand the effects of the two interventions on recycling behaviour, we carried out a naive direct regression on the contaminant levels on Phase 1 vs 2, and Phase 1 vs 3 data.

$$ContaminantType_{i,t} = \beta_0 + \beta_1 Treatment_i + \beta_2 Post_t + \beta_3 (Treatment_i \times Post_t) + \epsilon_{i,t}$$

Where

- $ContaminantType_{i,t}$ = Contaminant rate for group i at time t (3 separate models representing each contaminant type)
- $Treatment_i$ = dummy variable representing Engine (0) and UTown (1)
- $Post_t$ = dummy variable representing whether our observation is pre-implementation (0) or post-implementation (1)
- $Treatment_i \times Post_t$ = dummy variable representing UTown group post-implementation
- ϵ is the error term in our model

We use the StatsModel library in Python for the regression. Performing regression as per the base model above, we obtain the OLS regression result for each contaminant type for shaped openings seen in figure 6 and for Informational banners seen in figure 7.

3.5 Interpretation of results:

For both regressions, we get a β_0 of 0.8695, 49.0448 and 13.7729 for paper, plastic and can contaminant rate respectively. This intercept represents the estimated contaminant rate of the control (Engine) group before intervention.

We also get a β_1 of 5.2286, 10.6883 and 21.6928 for paper, plastic and can contaminant rate respectively. The β_1 coefficient represents the difference in contaminant rate between our treatment and control group before intervention (baseline bias). We see that our p-value is significant (p-value < 0.05) across all contaminant types indicating that UTown has a significant and higher contamination than their Engine counterparts across all three materials.

β_2 represents the difference in contaminant rate before and after the treatment, this is our indicator of the natural changes in contaminant rate over time. In DiD analysis, we use the parallel trends assumption to equate this to the natural changes in contaminant rate over time in the treatment group if it had not undergone the treatment. We notice in both models that only PlasticContaminant had a significant β_2 of -13.0869 and -9.6510 (both p-value < 0.05) indicating there is a significant decrease in plastic contaminant rate naturally with time.

β_3 coefficients	Effect of shaped openings (Phase I vs Phase II)		Effect of informational banners (Phase I vs Phase III)	
Material	Coefficient	Significant? (5%)	Coefficient	Significant? (5%)
Paper	-2.52	Yes	-1.30	No
Plastic	-0.139	No	+9.10	No
Can	-15.0	Yes	-7.24	No

Table 1: Values from the coefficient of interest quantifying the DID of different materials

β_3 is the Difference-in-Differences estimator and our coefficient of interest. It captures the additional change in contaminant rate for the treatment group post-treatment, relative to the control group. In our analysis of shaped openings, our β_3 is -2.5234 (p-value 0.048) and -15.0244 (0.007) for PaperContaminant and CanContaminant respectively indicating that introduction of shaped openings are effective in guiding users toward reducing contamination for paper and can recycling. Whereas PlasticContaminant with -0.1387 (p-value 0.978), we do not reject the null hypothesis that shaped openings do not significantly reduce PlasticContaminant. Although the decrease seen in PaperContaminant and CanContaminant could also have been due to pre-treatment diverging trends.

In our analysis of informational banners we see a starkly different effect, our β_3 is -1.3046 (p-value 0.288), and -7.2431 (p-value 0.133) for PaperContaminant and CanContaminant respectively indicates that introduction of information banners did not significantly reduce the PaperContaminant and CanContaminant. Moreover we also see a positive PlasticContaminant of +9.0998 (p-value 0.058) with introduction of informational banners, although this result was ultimately not significant as well. We can conclude that the use of informational banners do not significantly help to alter recycling behaviour of users, and any change in contamination levels from pre-treatment levels could be due to other external factors or noise.

3.6 DID Regression with fixed effects

We employ a **fixed-effects DID regression** that compares each unit to itself over time. This approach controls for unobserved, time-specific fixed effects by treating each unit as its own control group. By doing so, we hope to mitigate the violation of the parallel trends assumption by isolating the within-unit impact of the informational banner treatment. We implemented this by introducing an additional term to capture the unobservable counterfactual effect: the expected behaviour of UTOWN had it not received the Phase 2 treatment.

After accounting for this bias, the estimated treatment magnitudes were reduced to -0.3133 for paper and -1.3487 for cans, while plastic contamination remained largely stable at +9.154. The attenuation in estimated effects suggests that our earlier model may have overstated the treatment impact. The largely unchanged plastic contamination level aligns with our prior findings, indicating that variations in plastic contamination are primarily driven by random noise or external environmental factors rather than by the interventions itself.

4. Discussion

We notice some flaws in the rollout design of the study which makes controlling for bias difficult.

Firstly, the parallel trends assumption was not satisfied in Phase 1. A preliminary literature search revealed that this study was part of the NUS Sustainability Strategic Plan 2017-2020 (NUS, 2017) which also included other sustainability projects conducted at various locations around campus. It is possible that these other projects could have introduced additional confounding effects that resulted in diverging trends at baseline. As we did not have data on other confounders, we did not include them in this project.

We reason that this could also be due to insufficient data points to sufficiently show a parallel trend between Engine and UTown, as well as to absorb weekly volatility in contaminant rates. We propose to lengthen the duration of the Phase 1 data collection, as well as to expand data collection to other areas which similarly did not see implementation of sustainability projects. This allows us to sufficiently establish a less biased baseline.

Secondly, we acknowledge the distinctive set up of our experiment which involves two different treatments followed up sequentially over 3 phases. Staggered Treatment Adoption (Wing et al.) assumes that once a unit is treated, it remains treated for all subsequent time periods. This means that simply comparing Phase 1 and Phase 3 to test for treatment effect of informational banners is likely flawed due to lingering effects of shaped openings treatment, despite a “washout” period between Phase 2 and Phase 3. We propose that future studies can instead randomize sites into three separate groups: Control group, Treatment group A (implementation of shaped openings only) and Treatment group B (implementation of informational banners).

If not possible, data could instead also have been collected at UTown during the washout period between Phase 2 and 3 to better evaluate if the use of shaped openings had resulted in any systematic recycling behavioural changes which persist after the intervention. This allows us to test the effect of each treatment separately which reduces confounding effects.

5. Conclusion

Overall, our analysis concludes that informational banners are an ineffectual intervention in changing recycling behaviour, as compared to shaped openings, which resulted in decreased contamination for paper and cans. This suggests that physical design interventions were more effective than informational interventions in reducing contamination, as users may subconsciously choose to ignore the banners, whereas a physical barrier cannot be ignored.

Furthermore, both interventions were ineffective in reducing plastic contamination. This may be attributed to plastics being more easily contaminated by residual substances such as bottled drinks and food waste, compared to paper and cans. This is supported by the higher baseline contamination levels for plastics during the pre-treatment phase at both locations. The effort required to clean or separate contaminated plastics is likely perceived as too inconvenient, and the interventions implemented were insufficient to motivate users to change their disposal habits for improved recycling outcomes.

Appendix

Trial phase	Location	Days of data (no.)	Paper contamination rate (%)		Plastic contamination rate (%)		Cans contamination rate (%)	
			Mean	SD	Mean	SD	Mean	SD
1	ENGINE	17	0.87	0.39	49.04	12.14	13.77	8.94
1	UTOWN	17	6.10	3.56	59.73	6.96	35.47	10.30
2	ENGINE	11	1.29	2.04	35.96	9.09	11.67	9.66
2	UTOWN	11	4.00	1.48	46.51	7.23	18.34	10.18
3	ENGINE	11	0.77	1.41	39.39	7.59	7.86	8.27
3	UTOWN	11	4.69	1.77	59.18	3.52	22.31	5.11

Figure 1: Summary table of contamination data after data cleaning

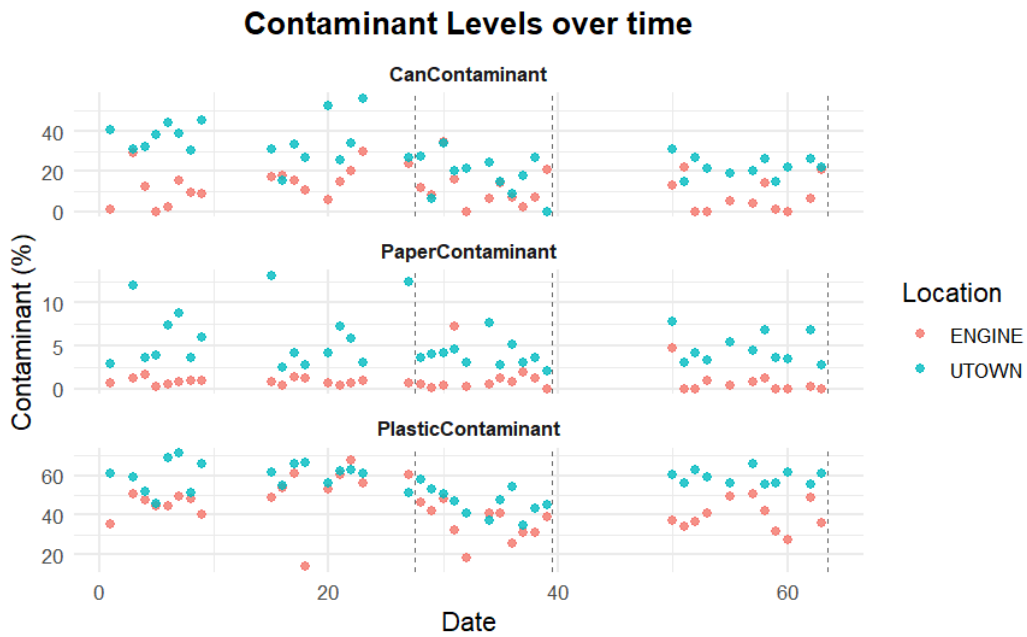


Figure 2: Scatter plot of contaminant levels over time. Boundary lines represent the separation between each of the three phases.

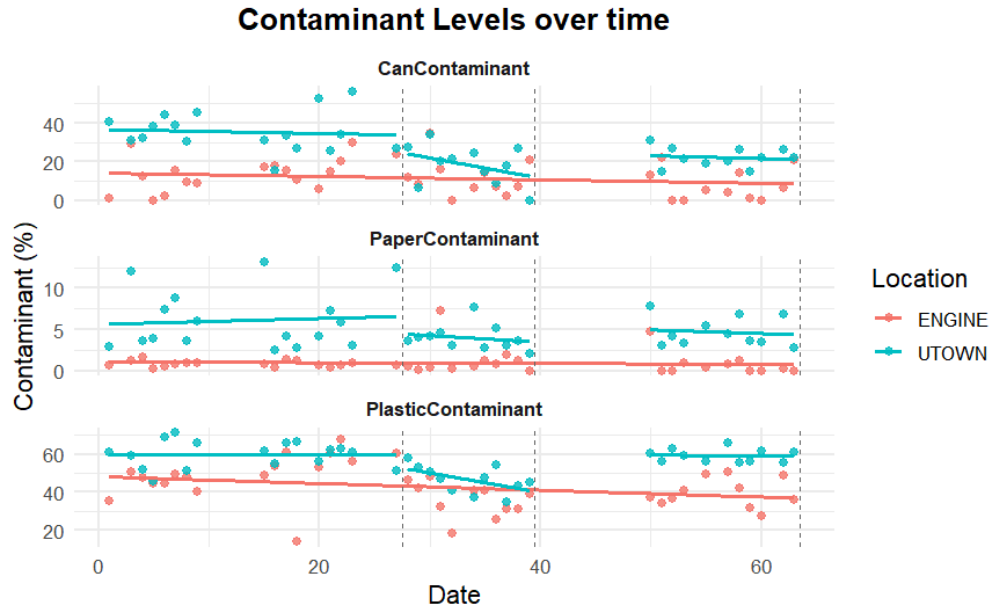


Figure 3: Contaminant levels of each material over time, with regression lines overlaid. Dotted lines represent the separation between each of the three phases. Engine (control) data was regressed over entire time period, while UTown (treatment) was regressed according to their respective phases.

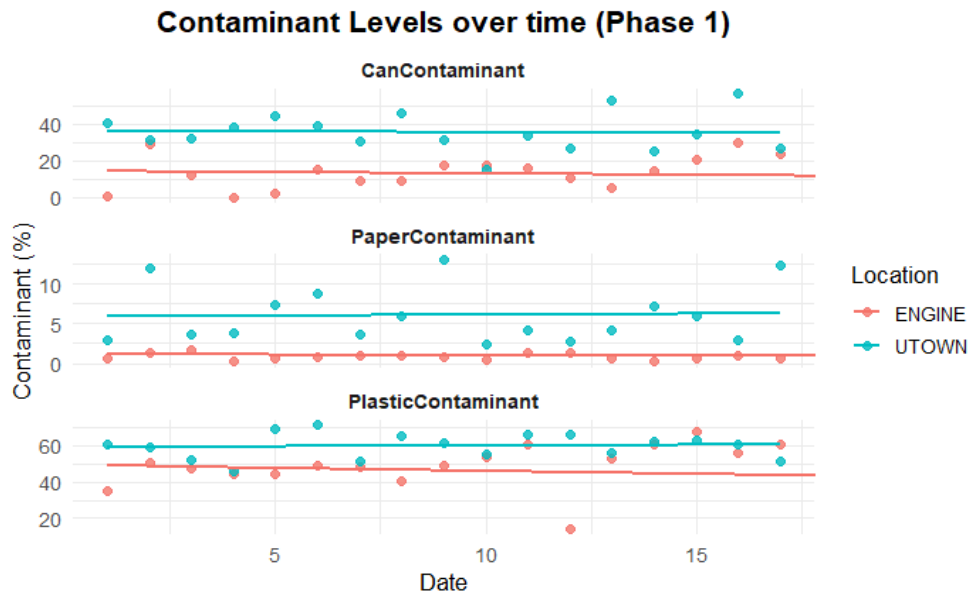


Figure 4: Contaminant rate of Paper, Plastic and Can in Phase 1

$$\text{ContaminantType} = \beta_0 + \beta_1 \text{Day} + \beta_2 (\text{Treatment} \times \text{Day}) + \epsilon$$

Where

- *ContaminantType* = Contaminant rate at given time point
- *Day* = day (in numerical value with 2020-01-15 as Day 1)
- *Treatment x Day* = 0 if Engine, Day if UTown

OLS Regression Results

Dep. Variable:	PaperContaminant	R-squared:	0.427			
Model:	OLS	Adj. R-squared:	0.390			
Method:	Least Squares	F-statistic:	11.53			
Date:	Sat, 18 Oct 2025	Prob (F-statistic):	0.000180			
Time:	00:49:18	Log-Likelihood:	-82.218			
No. Observations:	34	AIC:	170.4			
Df Residuals:	31	BIC:	175.0			
Df Model:	2					
Covariance Type:	nonrobust					

	coef	std err	t	P> t	[0.025	0.975]
Intercept	3.4398	1.020	3.371	0.002	1.359	5.521
Day	-0.2237	0.110	-2.027	0.051	-0.449	0.001
time_treatment	0.4573	0.095	4.802	0.000	0.263	0.651

OLS Regression Results

Dep. Variable:	PlasticContaminant	R-squared:	0.167			
Model:	OLS	Adj. R-squared:	0.113			
Method:	Least Squares	F-statistic:	3.098			
Date:	Sat, 18 Oct 2025	Prob (F-statistic):	0.0594			
Time:	00:59:05	Log-Likelihood:	-126.64			
No. Observations:	34	AIC:	259.3			
Df Residuals:	31	BIC:	263.9			
Df Model:	2					
Covariance Type:	nonrobust					

	coef	std err	t	P> t	[0.025	0.975]
Intercept	49.5286	3.769	13.142	0.000	41.842	57.215
Day	0.1867	0.408	0.458	0.650	-0.645	1.018
time_treatment	0.7068	0.352	2.010	0.053	-0.010	1.424

OLS Regression Results

Dep. Variable:	CanContaminant	R-squared:	0.373			
Model:	OLS	Adj. R-squared:	0.333			
Method:	Least Squares	F-statistic:	9.239			
Date:	Sat, 18 Oct 2025	Prob (F-statistic):	0.000713			
Time:	00:59:08	Log-Likelihood:	-130.80			
No. Observations:	34	AIC:	267.6			
Df Residuals:	31	BIC:	272.2			
Df Model:	2					
Covariance Type:	nonrobust					

	coef	std err	t	P> t	[0.025	0.975]
Intercept	21.0625	4.259	4.945	0.000	12.376	29.749
Day	-0.4378	0.461	-0.950	0.349	-1.377	0.502
time_treatment	1.6661	0.397	4.192	0.000	0.855	2.477

Figure 5: Statistical Test for Pre-Treatment Trend Differences Equation and the corresponding OLS Regression Results

OLS Regression Results						
Dep. Variable:	PaperContaminant	R-squared:				0.512
Model:	OLS	Adj. R-squared:				0.484
Method:	Least Squares	F-statistic:				18.19
Date:	Thu, 16 Oct 2025	Prob (F-statistic):				3.37e-08
Time:	00:05:46	Log-Likelihood:				-123.38
No. Observations:	56	AIC:				254.8
Df Residuals:	52	BIC:				262.9
Df Model:	3					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.8695	0.551	1.577	0.121	-0.237	1.976
treatment	5.2286	0.780	6.705	0.000	3.664	6.793
post	0.4253	0.880	0.484	0.631	-1.340	2.191
treatment_post	-2.5234	1.244	-2.028	0.048	-5.020	-0.027

OLS Regression Results						
Dep. Variable:	PlasticContaminant	R-squared:				0.465
Model:	OLS	Adj. R-squared:				0.434
Method:	Least Squares	F-statistic:				15.06
Date:	Thu, 16 Oct 2025	Prob (F-statistic):				3.53e-07
Time:	00:07:14	Log-Likelihood:				-202.17
No. Observations:	56	AIC:				412.3
Df Residuals:	52	BIC:				420.5
Df Model:	3					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	49.0448	2.252	21.780	0.000	44.526	53.564
treatment	10.6883	3.185	3.356	0.001	4.298	17.079
post	-13.0869	3.593	-3.643	0.001	-20.296	-5.877
treatment_post	-0.1387	5.081	-0.027	0.978	-10.334	10.057

OLS Regression Results						
Dep. Variable:	CanContaminant	R-squared:				0.526
Model:	OLS	Adj. R-squared:				0.498
Method:	Least Squares	F-statistic:				19.20
Date:	Thu, 16 Oct 2025	Prob (F-statistic):				1.64e-08
Time:	00:07:46	Log-Likelihood:				-204.92
No. Observations:	56	AIC:				417.8
Df Residuals:	52	BIC:				425.9
Df Model:	3					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	13.7729	2.365	5.823	0.000	9.027	18.519
treatment	21.6928	3.345	6.485	0.000	14.981	28.405
post	-2.0983	3.774	-0.556	0.581	-9.670	5.474
treatment_post	-15.0244	5.337	-2.815	0.007	-25.733	-4.316

PaperContaminant
 $0.8695 + 5.2286\text{treatment} + 0.4253\text{post} - 2.5234\text{treatment_post} + \epsilon$

PlasticContaminant
 $49.0448 + 10.6883\text{treatment} - 13.0869\text{post} - 0.1387\text{treatment_post} + \epsilon$

CanContaminant
 $13.7729 + 21.6928\text{treatment} - 2.0983\text{post} - 15.0244\text{treatment_post} + \epsilon$

Figure 6: DID Regression for PaperContaminant, PlasticContaminant and CanContaminant for Shaped Openings

OLS Regression Results						
=====						
Dep. Variable:	PaperContaminant	R-squared:	0.559			
Model:	OLS	Adj. R-squared:	0.533			
Method:	Least Squares	F-statistic:	21.96			
Date:	Thu, 16 Oct 2025	Prob (F-statistic):	2.54e-09			
Time:	08:46:52	Log-Likelihood:	-122.05			
No. Observations:	56	AIC:	252.1			
Df Residuals:	52	BIC:	260.2			
Df Model:	3					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	0.8695	0.538	1.615	0.112	-0.211	1.950
treatment	5.2286	0.761	6.867	0.000	3.701	6.757
post	-0.1028	0.859	-0.120	0.905	-1.827	1.621
treatment_post	-1.3046	1.215	-1.074	0.288	-3.742	1.133
=====						

OLS Regression Results						
Dep. Variable:	PlasticContaminant	R-squared:	0.475			
Model:	OLS	Adj. R-squared:	0.445			
Method:	Least Squares	F-statistic:	15.70			
Date:	Thu, 16 Oct 2025	Prob (F-statistic):	2.15e-07			
Time:	08:47:11	Log-Likelihood:	-197.80			
No. Observations:	56	AIC:	403.6			
Df Residuals:	52	BIC:	411.7			
Df Model:	3					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	49.0448	2.082	23.551	0.000	44.866	53.224
treatment	10.6883	2.945	3.629	0.001	4.779	16.598
post	-9.6510	3.323	-2.905	0.005	-16.318	-2.984
treatment_post	9.0998	4.699	1.937	0.058	-0.329	18.529

OLS Regression Results						
=====						
Dep. Variable:	CanContaminant	R-squared:	0.619			
Model:	OLS	Adj. R-squared:	0.597			
Method:	Least Squares	F-statistic:	28.14			
Date:	Thu, 16 Oct 2025	Prob (F-statistic):	5.98e-11			
Time:	08:47:14	Log-Likelihood:	-198.42			
No. Observations:	56	AIC:	404.8			
Df Residuals:	52	BIC:	412.9			
Df Model:	3					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	13.7729	2.106	6.541	0.000	9.547	17.998
treatment	21.6928	2.978	7.284	0.000	15.717	27.669
post	-5.9166	3.360	-1.761	0.084	-12.658	0.825
treatment_post	-7.2431	4.751	-1.524	0.133	-16.777	2.291

PaperContaminant
 $0.8695 + 5.2286\text{treatment} - 0.1028\text{post} - 1.3046\text{treatment_post} + \varepsilon$

PlasticContaminant
 $49.0448 + 10.6883\text{treatment} - 9.6510\text{post} + 9.0998\text{treatment_post} + \varepsilon$

CanContaminant
 $13.7729 + 21.6928\text{treatment} - 5.9166\text{post} - 7.2431\text{treatment_post} + \varepsilon$

Figure 7: DID Regression for PaperContaminant, PlasticContaminant and CanContaminant for Informational Banners

OLS Regression Results						
Dep. Variable:	PaperContaminant	R-squared:	0.505			
Model:	OLS	Adj. R-squared:	0.478			
Method:	Least Squares	F-statistic:	18.63			
Date:	Sat, 18 Oct 2025	Prob (F-statistic):	1.37e-10			
Time:	15:51:47	Log-Likelihood:	-168.23			
No. Observations:	78	AIC:	346.5			
Df Residuals:	73	BIC:	358.2			
Df Model:	4					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	1.3651	0.470	2.904	0.005	0.428	2.302
C(FirstTrialPhase)[T.2.0]	-0.8364	0.592	-1.414	0.162	-2.015	0.343
C(FirstTrialPhase)[T.3.0]	-0.5985	0.804	-0.745	0.459	-2.200	1.003
treatment	4.2372	0.578	7.334	0.000	3.086	5.389
treatment_post	-0.3133	1.088	-0.288	0.774	-2.481	1.855
OLS Regression Results						
Dep. Variable:	CanContaminant	R-squared:	0.491			
Model:	OLS	Adj. R-squared:	0.463			
Method:	Least Squares	F-statistic:	17.57			
Date:	Sat, 18 Oct 2025	Prob (F-statistic):	3.85e-10			
Time:	15:51:47	Log-Likelihood:	-284.02			
No. Observations:	78	AIC:	578.0			
Df Residuals:	73	BIC:	589.8			
Df Model:	4					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	16.7241	2.074	8.063	0.000	12.590	20.858
C(FirstTrialPhase)[T.2.0]	-9.6105	2.610	-3.682	0.000	-14.813	-4.408
C(FirstTrialPhase)[T.3.0]	-8.8678	3.546	-2.501	0.015	-15.935	-1.800
treatment	15.7903	2.550	6.193	0.000	10.709	20.872
treatment_post	-1.3407	4.801	-0.279	0.781	-10.909	8.227
OLS Regression Results						
Dep. Variable:	PlasticContaminant	R-squared:	0.539			
Model:	OLS	Adj. R-squared:	0.513			
Method:	Least Squares	F-statistic:	21.30			
Date:	Sat, 18 Oct 2025	Prob (F-statistic):	1.13e-11			
Time:	15:51:47	Log-Likelihood:	-274.33			
No. Observations:	78	AIC:	558.7			
Df Residuals:	73	BIC:	570.4			
Df Model:	4					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	49.0720	1.832	26.789	0.000	45.421	52.723
C(FirstTrialPhase)[T.2.0]	-13.1562	2.305	-5.707	0.000	-17.751	-8.562
C(FirstTrialPhase)[T.3.0]	-9.6782	3.132	-3.090	0.003	-15.920	-3.436
treatment	10.6338	2.252	4.722	0.000	6.146	15.122
treatment_post	9.1543	4.240	2.159	0.034	0.704	17.605
aminant						
- 0.8364C(FirstTrialPhase)[T.2.0] - 0.5985C(FirstTrialPhase)[T.3.0] + 4.2372treatment - 0.3133treatment_post + \epsilon						
ntaminant						
- 13.1562C(FirstTrialPhase)[T.2.0] - 9.6782C(FirstTrialPhase)[T.3.0] + 10.6338treatment + 9.1543treatment_post + \epsilon						
inant						
- 9.6105C(FirstTrialPhase)[T.2.0] - 8.8678C(FirstTrialPhase)[T.3.0] + 15.7903treatment - 1.3407treatment_post + \epsilon						

Figure 8: DID with FE Regression for PaperContaminant, PlasticContaminant and CanContaminant for Informational Banners

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